

STAT 423 Winter 2026 - Final Project

Krish Doshi, Lauren Yee, Kian Sharifzadeh

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Data Processing - Krish Doshi

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating,
  age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified",
  orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
  ifelse(grepl("Country", orig_data$cdr_genre), "Country",
  ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
  ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
  ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy,
  danceability = orig_data$danceability, happiness = orig_data$happiness,
  loudness_db = orig_data$loudness_d_b,
  length_sec = orig_data$length_sec)

cleaned_data$year <- as.numeric(substr(orig_data$date, 1,
  unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

model_results <- list()

for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- subpop_60s <- cleaned_data[cleaned_data$year >= i - 10 &
    cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
```

```

if(nrow(subpop) < 50) {
  message(paste("Skipping", decade_label, "- Not enough data points."))
  next
}

factor_counts <- sapply(subpop[, c("genre", "gender", "race")],
  function(x) length(unique(na.omit(x))))

if(any(factor_counts < 2)) {
  message(paste("Skipping", decade_label, "due to single-level factor:",
    names(factor_counts)[factor_counts < 2]))
  next
}

first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy
  + danceability + happiness + bpm + loudness_db + length_sec)

second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre
  + age*energy + age*danceability + age*happiness + age*bpm
  + age*loudness_db + age*length_sec + race*genre + race*energy
  + race*danceability + race*happiness + race*bpm + race*loudness_db
  + race*length_sec + gender*genre + gender*energy + gender*danceability
  + gender*happiness + gender*bpm + gender*loudness_db + gender*length_sec )

# --- Plotting (Wrapped in 'try' so one bad plot doesn't stop the whole loop) ---
# try({
#   plot(first_model, which = 1:2)
#   boxcox(first_model, plotit=TRUE)
#   plot(second_model, which = 1:2)
#   boxcox(second_model, plotit=TRUE)
# })
optimal_model <- step(second_model, direction="backward")

model_results[[decade_label]] <- list(
  "first" = first_model,
  "second" = second_model,
  "optimal" = optimal_model
)

print(decade_label)

summary(first_model)

summary(second_model)

summary(optimal_model)
}

## Skipping 1950s - Not enough data points.
## Skipping 2020s due to single-level factor: genre
# Loop through each decade in the model_results list
for (decade in names(model_results)) {

```

```

# Check if the 'optimal' model exists for this decade
if (!is.null(model_results[[decade]]$optimal)) {

  cat("\n", "=====", "\n")
  cat(" SUMMARY FOR OPTIMAL MODEL:", decade, "\n")
  cat("=====", "\n")

  # Print the summary
  print(summary(model_results[[decade]]$optimal))

} else {
  cat("\nNo optimal model found for:", decade, "\n")
}
}

##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1960s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.3064 -1.2960  0.1313  1.2436  4.2045
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.235892   2.001360   3.116  0.00214 **
## age             -0.036813   0.021122  -1.743  0.08309 .
## genderNot Male   -1.820687   1.293523  -1.408  0.16101
## raceWhite        1.604179   1.795663   0.893  0.37287
## genreJazz        1.473039   1.332354   1.106  0.27040
## genreOther       1.978211   0.999147   1.980  0.04926 *
## genrePop         0.988531   0.889417   1.111  0.26788
## genreRock        0.260014   0.629236   0.413  0.67994
## energy          -0.005765   0.015336  -0.376  0.70743
## danceability     -0.013030   0.022465  -0.580  0.56264
## happiness        0.004845   0.009013   0.538  0.59154
## length_sec       0.008965   0.003791   2.365  0.01912 *
## raceWhite:genreJazz -1.353373   2.222521  -0.609  0.54334
## raceWhite:genreOther -3.249204   1.187123  -2.737  0.00683 **
## raceWhite:genrePop  -1.786329   0.784348  -2.277  0.02395 *
## raceWhite:genreRock      NA         NA      NA      NA
## raceWhite:energy      0.024399   0.017347   1.407  0.16132
## raceWhite:danceability -0.046982   0.022464  -2.091  0.03791 *
## genderNot Male:danceability 0.076584   0.023552   3.252  0.00137 **
## genderNot Male:happiness -0.030694   0.016106  -1.906  0.05830 .
## ---

```

```

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.797 on 178 degrees of freedom
## Multiple R-squared:  0.314, Adjusted R-squared:  0.2446
## F-statistic: 4.527 on 18 and 178 DF,  p-value: 4.551e-08
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1970s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:happiness + age:bpm + age:length_sec + race:energy +
##     gender:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.1773 -1.0450  0.0378  1.0913  4.6561
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.7599806   6.0727470   0.784  0.43397
## age              0.0742298   0.1986635   0.374  0.70902
## genderNot Male   -3.3233372   1.3575270  -2.448  0.01513 *
## raceWhite       -1.5914109   0.7655701  -2.079  0.03878 *
## genreJazz        2.9396820   2.5394662   1.158  0.24826
## genreOther      -5.4187148   4.2740093  -1.268  0.20618
## genrePop        -3.9847287   4.0475549  -0.984  0.32594
## genreRock       -6.6280029   4.3994636  -1.507  0.13334
## energy          -0.0217547   0.0131035  -1.660  0.09827 .
## happiness        0.0681171   0.0256047   2.660  0.00837 **
## bpm             -0.0531808   0.0330717  -1.608  0.10923
## loudness_db      0.0702293   0.0514647   1.365  0.17374
## length_sec       0.0505438   0.0166865   3.029  0.00274 **
## age:genreJazz      NA         NA         NA     NA
## age:genreOther     0.2154805   0.1350105   1.596  0.11189
## age:genrePop       0.1145362   0.1272669   0.900  0.36910
## age:genreRock      0.2268970   0.1398953   1.622  0.10623
## age:happiness     -0.0020628   0.0008623  -2.392  0.01758 *
## age:bpm           0.0017767   0.0011320   1.570  0.11794
## age:length_sec    -0.0015432   0.0005939  -2.599  0.00998 **
## raceWhite:energy   0.0207272   0.0131130   1.581  0.11537
## genderNot Male:length_sec 0.0151038   0.0060578   2.493  0.01338 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.772 on 224 degrees of freedom
## Multiple R-squared:  0.2958, Adjusted R-squared:  0.233
## F-statistic: 4.705 on 20 and 224 DF,  p-value: 2.095e-09
##
##
## =====

```

```
## SUMMARY FOR OPTIMAL MODEL: 1980s
## =====
##
## Call:
## lm(formula = rating ~ race + genre + bpm + length_sec + race:genre +
##     race:bpm + race:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4534 -0.8562  0.0485  0.9861  3.9808
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.289636    1.681818   1.361  0.1748
## raceWhite         2.987554    1.951337   1.531  0.1272
## genrePop        -0.726606    0.413305  -1.758  0.0801 .
## genreRock       -0.153197    0.492855  -0.311  0.7562
## bpm              0.011539    0.007813   1.477  0.1412
## length_sec       0.011023    0.005228   2.108  0.0361 *
## raceWhite:genrePop  1.329615    0.537348   2.474  0.0141 *
## raceWhite:genreRock  0.819251    0.591083   1.386  0.1672
## raceWhite:bpm     -0.016139    0.009210  -1.752  0.0811 .
## raceWhite:length_sec -0.009072    0.006014  -1.509  0.1329
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.492 on 218 degrees of freedom
## Multiple R-squared:  0.06512,    Adjusted R-squared:  0.02652
## F-statistic: 1.687 on 9 and 218 DF,  p-value: 0.09338
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1990s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + loudness_db + length_sec + age:energy +
##     age:danceability + age:loudness_db + race:danceability +
##     race:length_sec + gender:genre + gender:danceability + gender:loudness_db +
##     gender:length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7557 -0.9125  0.0000  0.7949  3.6288
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      10.151452    4.862418   2.088 0.039069 *
## age             -0.188701    0.136443  -1.383 0.169390
## genderNot Male     4.498917    3.029352   1.485 0.140299
## raceWhite         5.519565    3.041628   1.815 0.072227 .
## genreOther       -0.174664    1.634218  -0.107 0.915074
## genrePop          0.268361    1.665154   0.161 0.872253
```

```

## genreRock          0.835016    1.632311    0.512 0.609961
## energy             -0.102502    0.032920   -3.114 0.002341 **
## danceability        0.119070    0.050887    2.340 0.021045 *
## happiness           0.017499    0.007628    2.294 0.023637 *
## loudness_db         1.147865    0.316756    3.624 0.000437 ***
## length_sec          0.011877    0.007043    1.686 0.094503 .
## age:energy           0.002489    0.001100    2.262 0.025630 *
## age:danceability    -0.003376    0.001633   -2.067 0.040979 *
## age:loudness_db     -0.028004    0.010459   -2.678 0.008522 **
## raceWhite:danceability -0.035957    0.024752   -1.453 0.149090
## raceWhite:length_sec -0.014350    0.008659   -1.657 0.100228
## genderNot Male:genreOther  1.606214    0.980833    1.638 0.104286
## genderNot Male:genrePop    0.660151    1.026215    0.643 0.521341
## genderNot Male:genreRock      NA          NA          NA          NA
## genderNot Male:danceability -0.037786    0.023108   -1.635 0.104790
## genderNot Male:loudness_db -0.199855    0.119747   -1.669 0.097891 .
## genderNot Male:length_sec  -0.015603    0.008158   -1.913 0.058336 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.429 on 113 degrees of freedom
## Multiple R-squared:  0.374, Adjusted R-squared:  0.2577
## F-statistic: 3.215 on 21 and 113 DF,  p-value: 3.432e-05
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2000s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.80175 -0.88847 -0.00231  1.02953  2.93881
##
## Coefficients: (2 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    14.254799    5.689019   2.506 0.013714 *
## age            -0.262392    0.164223  -1.598 0.113013
## genderNot Male  -0.551642    2.208098  -0.250 0.803196
## raceWhite      -13.479191    3.724145  -3.619 0.000451 ***
## genreOther       0.362481    1.668695   0.217 0.828443
## genrePop         0.218854    1.608297   0.136 0.892013
## genreRock        0.090807    1.604975   0.057 0.954986
## energy           0.018882    0.011341   1.665 0.098826 .
## danceability    -0.083852    0.065397  -1.282 0.202519
## happiness        0.001354    0.011393   0.119 0.905583
## bpm             -0.014549    0.008120  -1.792 0.075985 .
## length_sec      -0.009048    0.005130  -1.764 0.080619 .

```

```

## age:danceability          0.003737    0.002195    1.703 0.091492 .
## raceWhite:genreOther     -1.186109    0.866090   -1.369 0.173685
## raceWhite:genrePop              NA          NA          NA    NA
## raceWhite:genreRock              NA          NA          NA    NA
## raceWhite:danceability     0.084078    0.031491    2.670 0.008759 **
## raceWhite:happiness       -0.025028    0.018899   -1.324 0.188199
## raceWhite:bpm              0.035295    0.012426    2.840 0.005385 **
## raceWhite:length_sec       0.019703    0.011520    1.710 0.090064 .
## genderNot Male:danceability -0.041970    0.024007   -1.748 0.083262 .
## genderNot Male:happiness    0.021226    0.014487    1.465 0.145771
## genderNot Male:bpm          0.019019    0.010619    1.791 0.076076 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.39 on 108 degrees of freedom
## Multiple R-squared:  0.3417, Adjusted R-squared:  0.2198
## F-statistic: 2.803 on 20 and 108 DF,  p-value: 0.0003367
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2010s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + genre + energy + happiness +
##     bpm + length_sec + age:genre + age:bpm + age:length_sec +
##     gender:genre + gender:energy + gender:happiness, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6153 -0.7497  0.0326  0.8805  3.4548
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.7479129   5.9555613   -0.126  0.90032
## age             0.2078850   0.2138097    0.972  0.33328
## genderNot Male -1.4345612   1.2782808   -1.122  0.26447
## genrePop        4.0067252   1.8431272    2.174  0.03210 *
## genreRock      -1.3841166   6.8130047   -0.203  0.83943
## energy         -0.0346294   0.0140445   -2.466  0.01540 *
## happiness       0.0176167   0.0096788    1.820  0.07176 .
## bpm           -0.0504318   0.0327145   -1.542  0.12637
## length_sec      0.0611063   0.0235217    2.598  0.01081 *
## age:genrePop   -0.1641477   0.0621883   -2.640  0.00965 **
## age:genreRock   0.0665065   0.2087839    0.319  0.75074
## age:bpm         0.0017263   0.0011530    1.497  0.13752
## age:length_sec -0.0017601   0.0008402   -2.095  0.03874 *
## genderNot Male:genrePop  1.5747664   0.7167070    2.197  0.03034 *
## genderNot Male:genreRock      NA          NA          NA    NA
## genderNot Male:energy    0.0408389   0.0215520    1.895  0.06102 .
## genderNot Male:happiness -0.0271834   0.0184388   -1.474  0.14359
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Residual standard error: 1.5 on 99 degrees of freedom
## Multiple R-squared:  0.2797, Adjusted R-squared:  0.1706
## F-statistic: 2.563 on 15 and 99 DF,  p-value: 0.002865
```

Preliminary Exploration of Data through Visual Analysis

Plots of Individual Relationships

The plots show the relationship between the predictors and the response variable, rating.

```
library(ggplot2)
library(patchwork)

##
## Attaching package: 'patchwork'
## The following object is masked from 'package:MASS':
##
##      area

plot_age <- ggplot(cleaned_data, aes(x = age, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Age")

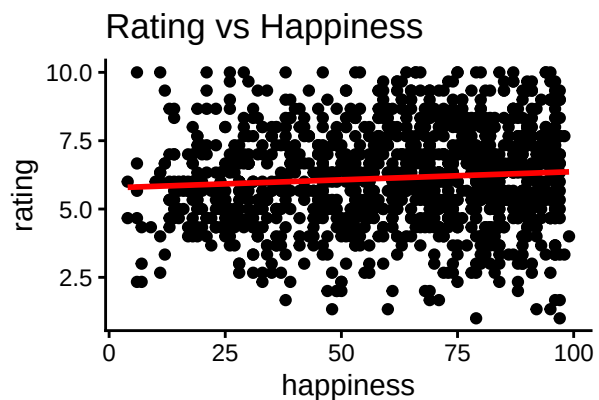
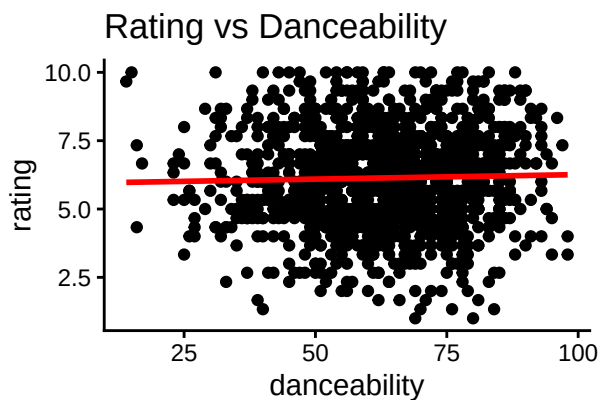
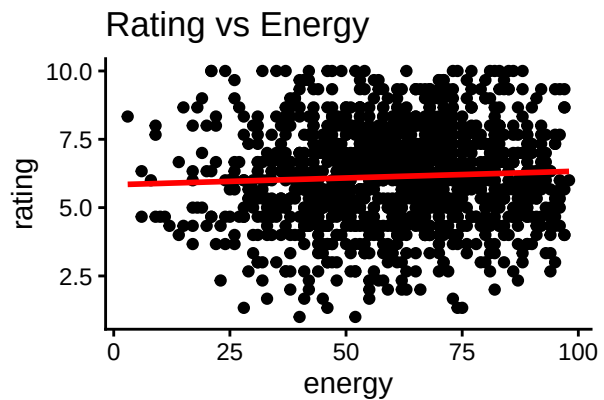
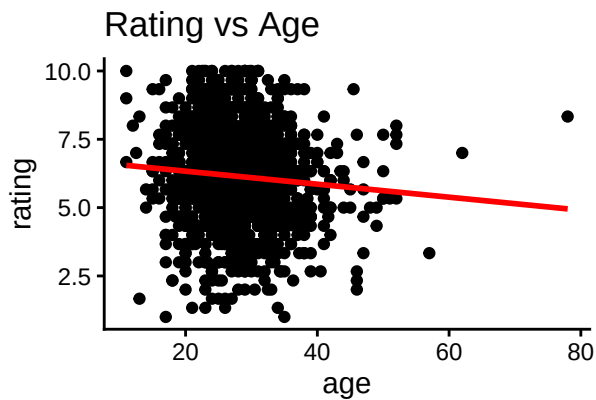
plot_energy <- ggplot(cleaned_data, aes(x = energy, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Energy")

plot_danceability <- ggplot(cleaned_data, aes(x = danceability, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Danceability")

plot_happiness <- ggplot(cleaned_data, aes(x = happiness, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Happiness")

plot_age + plot_energy + plot_danceability + plot_happiness

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

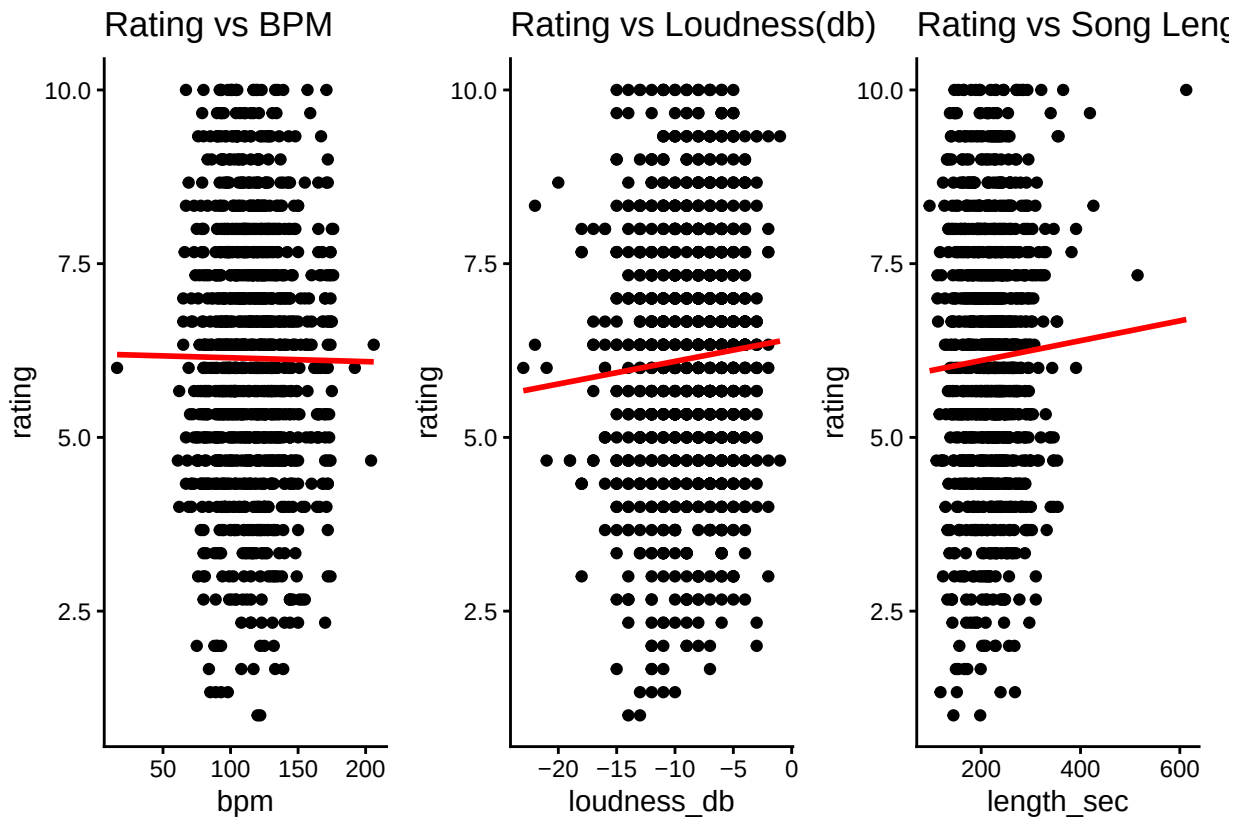
```
plot_bpm <- ggplot(cleaned_data, aes(x = bpm, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs BPM")

plot_loudness_db <- ggplot(cleaned_data, aes(x = loudness_db, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Loudness(db)")

plot_length_sec <- ggplot(cleaned_data, aes(x = length_sec, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Song Length(sec)")

plot_bpm + plot_loudness_db + plot_length_sec

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



```

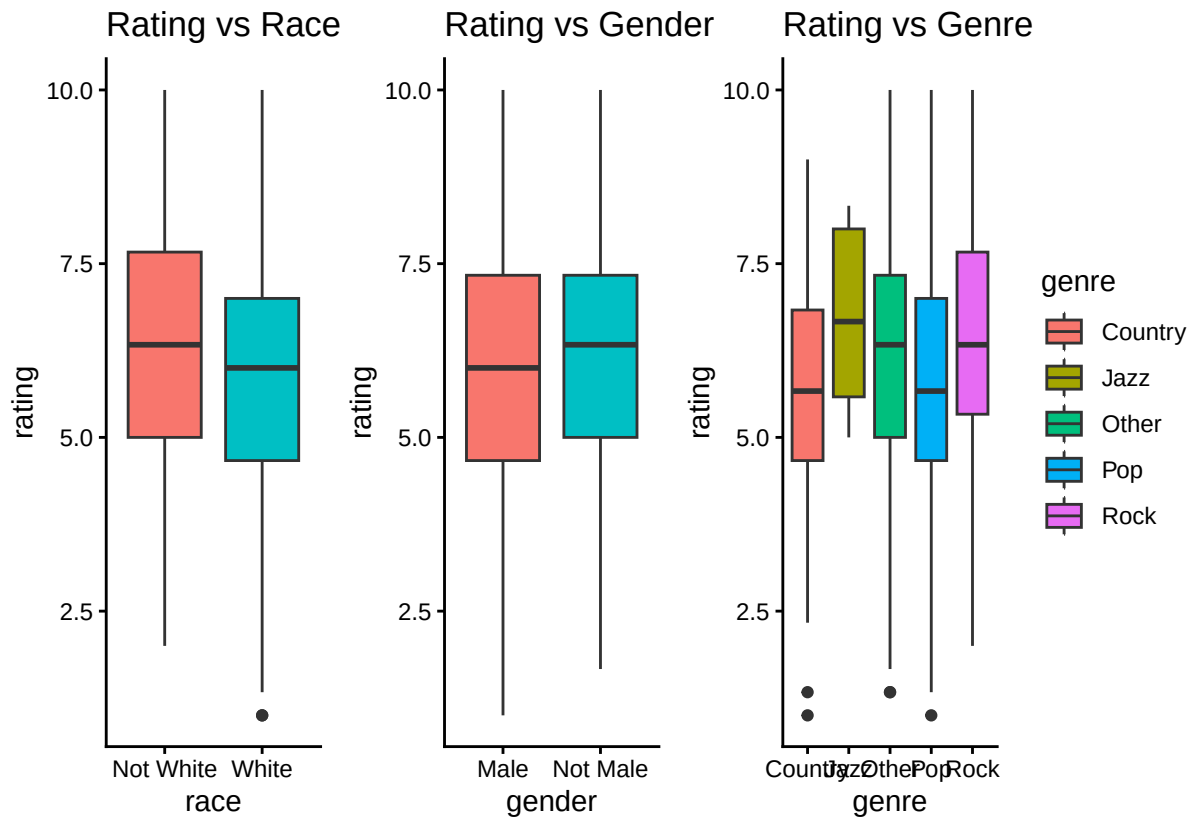
plot_race <- ggplot(cleaned_data, aes(x = race, y = rating, fill = race)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Race") +
  guides(fill = "none")

plot_gender <- ggplot(cleaned_data, aes(x = gender, y = rating, fill = gender)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Gender") +
  guides(fill = "none")

plot_genre <- ggplot(cleaned_data, aes(x = genre, y = rating, fill = genre)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Genre")

plot_race + plot_gender + plot_genre

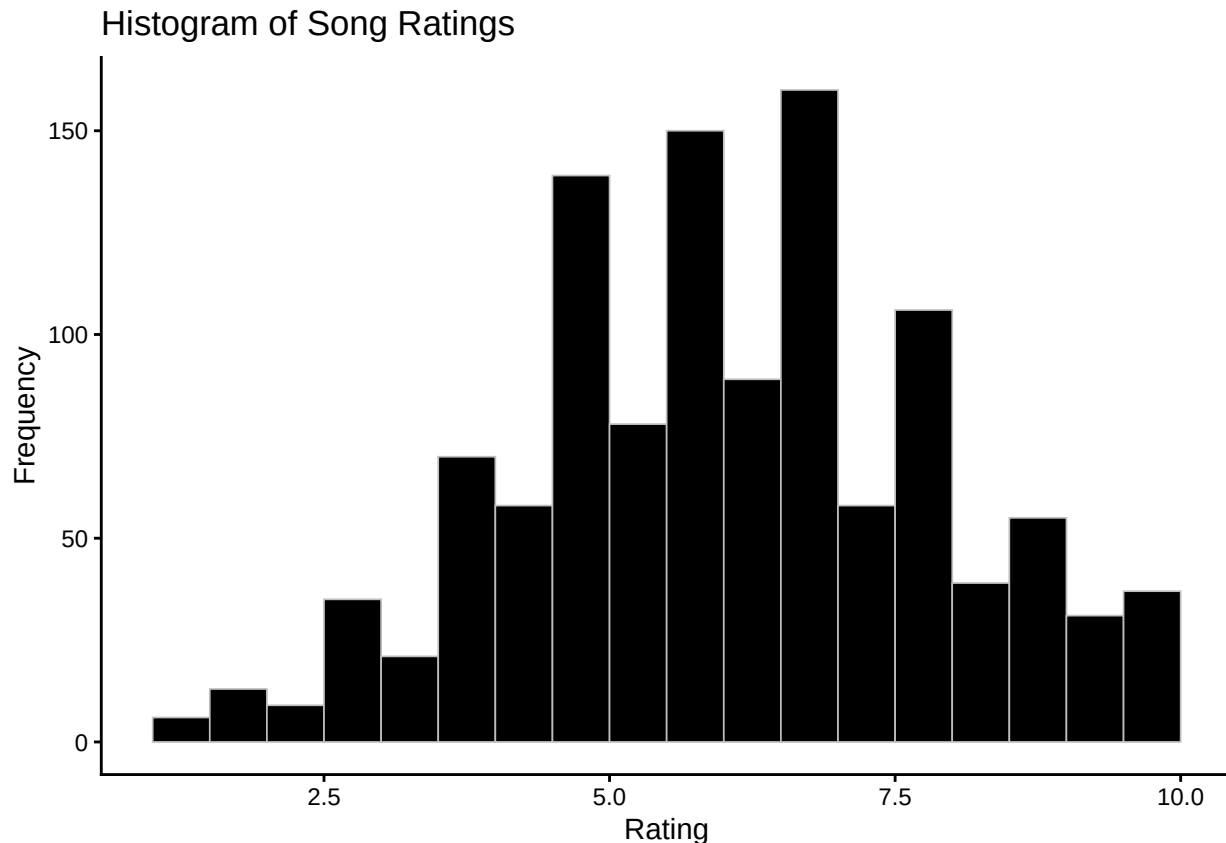
```



As

expected, banding is observed in the scatterplots because the response variable, rating, is discrete. Since race, gender, and genre are categorical predictors, the relationship with rating is displayed by boxplots instead.

```
ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 0.5,
    boundary = 0.5,
    fill = "black",
    color = "grey",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()
```



However, the histogram of ratings appears approximately symmetric and unimodal, resembling a normal distribution. While the ratings are discrete (integer-valued) and there are many categorical predictors, the distribution spans multiple levels, creating a shape that approximates continuity. This supports the treating of rating as a continuous response variable in the regression analysis.

Plots of Grouped Relationships

Data visualizations for grouped relationships help us examine whether relationships between predictors and ratings differ across categories. These plots justify the inclusion of categorical predictors in the regression models and help identify potential interaction effects.

```
model_60s <- model_results[["1960s"]]$optimal
model_70s <- model_results[["1970s"]]$optimal
model_80s <- model_results[["1980s"]]$optimal
model_90s <- model_results[["1990s"]]$optimal
model_2000s <- model_results[["2000s"]]$optimal
model_2010s <- model_results[["2010s"]]$optimal

vars_by_decade <- lapply(model_results, function(x) {
  attr(terms(x$optimal), "term.labels")
})

all_vars <- unique(unlist(vars_by_decade))

presence_matrix <- sapply(vars_by_decade, function(vars) {
  all_vars %in% vars
})
```

```
rownames(presence_matrix) <- all_vars
presence_matrix
```

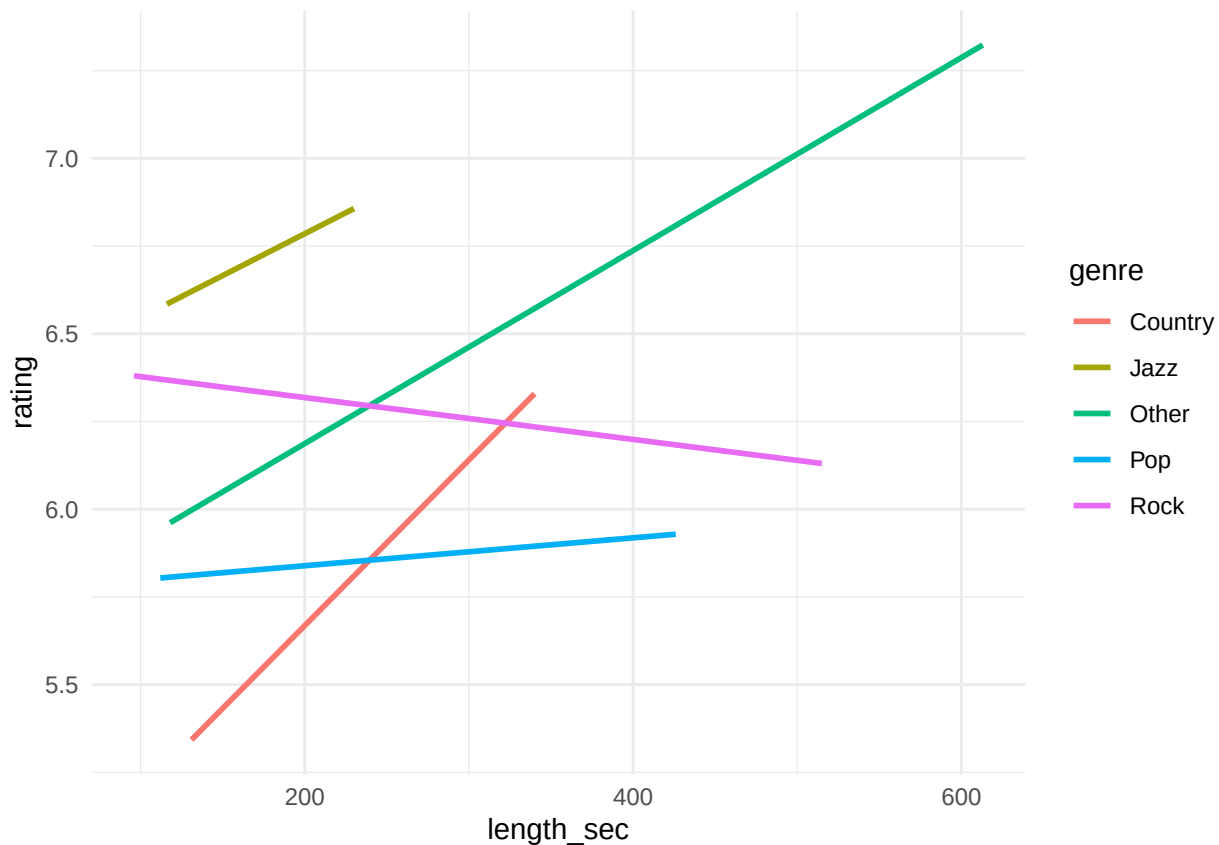
```
##           1960s 1970s 1980s 1990s 2000s 2010s
## age          TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## gender        TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## race          TRUE  TRUE  TRUE  TRUE  TRUE FALSE
## genre         TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## energy        TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## danceability  TRUE FALSE FALSE  TRUE  TRUE FALSE
## happiness     TRUE  TRUE FALSE  TRUE  TRUE  TRUE
## length_sec    TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## race:genre    TRUE FALSE  TRUE FALSE  TRUE FALSE
## race:energy   TRUE  TRUE FALSE FALSE FALSE FALSE
## race:danceability TRUE FALSE FALSE  TRUE  TRUE FALSE
## gender:danceability TRUE FALSE FALSE  TRUE  TRUE FALSE
## gender:happiness TRUE FALSE FALSE FALSE  TRUE  TRUE
## bpm          FALSE  TRUE  TRUE FALSE  TRUE  TRUE
## loudness_db   FALSE  TRUE FALSE  TRUE FALSE FALSE
## age:genre     FALSE  TRUE FALSE FALSE FALSE  TRUE
## age:happiness FALSE  TRUE FALSE FALSE FALSE FALSE
## age:bpm       FALSE  TRUE FALSE FALSE FALSE  TRUE
## age:length_sec FALSE  TRUE FALSE FALSE FALSE  TRUE
## gender:length_sec FALSE  TRUE FALSE  TRUE FALSE FALSE
## race:bpm      FALSE FALSE  TRUE FALSE  TRUE FALSE
## race:length_sec FALSE FALSE  TRUE  TRUE  TRUE FALSE
## age:energy    FALSE FALSE FALSE  TRUE FALSE FALSE
## age:danceability FALSE FALSE FALSE  TRUE  TRUE FALSE
## age:loudness_db FALSE FALSE FALSE  TRUE FALSE FALSE
## gender:genre  FALSE FALSE FALSE  TRUE FALSE  TRUE
## gender:loudness_db FALSE FALSE FALSE  TRUE FALSE FALSE
## race:happiness FALSE FALSE FALSE FALSE  TRUE FALSE
## gender:bpm    FALSE FALSE FALSE FALSE  TRUE FALSE
## gender:energy FALSE FALSE FALSE FALSE FALSE  TRUE
```

```
rownames(presence_matrix)[rowSums(presence_matrix) == ncol(presence_matrix)]
```

```
## [1] "genre"      "length_sec"
```

```
ggplot(cleaned_data, aes(x = length_sec, y = rating)) +
  geom_smooth(aes(color = genre), method = "lm", se = FALSE) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The fitted regression lines indicate that the relationship between song length and rating varies by genre. In particular, certain genres (e.g., Other and Country) exhibit steeper positive slopes, while others (e.g., Rock and Pop) show weaker or slightly negative trends, suggesting a potential interaction between length and genre.

model_60s

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
##
## Coefficients:
##              (Intercept)              age
##                6.235892             -0.036813
##      genderNot Male      raceWhite
##        -1.820687              1.604179
##      genreJazz      genreOther
##         1.473039             1.978211
##      genrePop      genreRock
##         0.988531             0.260014
##         energy      danceability
##        -0.005765            -0.013030
##        happiness      length_sec
##         0.004845             0.008965
##      raceWhite:genreJazz      raceWhite:genreOther
```

```
##           -1.353373           -3.249204
##           raceWhite:genrePop       raceWhite:genreRock
##           -1.786329                NA
##           raceWhite:energy       raceWhite:danceability
##           0.024399                -0.046982
## genderNot Male:danceability       genderNot Male:happiness
##           0.076584                -0.030694
```

model_70s

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     happiness + bpm + loudness_db + length_sec + age:genre +
##     age:happiness + age:bpm + age:length_sec + race:energy +
##     gender:length_sec, data = subpop)
##
## Coefficients:
##           (Intercept)                age
##           4.759981                0.074230
##           genderNot Male                raceWhite
##           -3.323337                -1.591411
##           genreJazz                genreOther
##           2.939682                -5.418715
##           genrePop                genreRock
##           -3.984729                -6.628003
##           energy                happiness
##           -0.021755                0.068117
##           bpm                loudness_db
##           -0.053181                0.070229
##           length_sec                age:genreJazz
##           0.050544                NA
##           age:genreOther                age:genrePop
##           0.215480                0.114536
##           age:genreRock                age:happiness
##           0.226897                -0.002063
##           age:bpm                age:length_sec
##           0.001777                -0.001543
##           raceWhite:energy       genderNot Male:length_sec
##           0.020727                0.015104
```

model_80s

```
##
## Call:
## lm(formula = rating ~ race + genre + bpm + length_sec + race:genre +
##     race:bpm + race:length_sec, data = subpop)
##
## Coefficients:
##           (Intercept)                raceWhite                genrePop
##           2.289636                2.987554                -0.726606
##           genreRock                bpm                length_sec
##           -0.153197                0.011539                0.011023
##           raceWhite:genrePop       raceWhite:genreRock       raceWhite:bpm
##           1.329615                0.819251                -0.016139
##           raceWhite:length_sec
```

```
##          -0.009072
model_90s

##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##      danceability + happiness + loudness_db + length_sec + age:energy +
##      age:danceability + age:loudness_db + race:danceability +
##      race:length_sec + gender:genre + gender:danceability + gender:loudness_db +
##      gender:length_sec, data = subpop)
##
## Coefficients:
##          (Intercept)                age
##          10.151452             -0.188701
##      genderNot Male             raceWhite
##          4.498917             5.519565
##      genreOther             genrePop
##      -0.174664             0.268361
##      genreRock             energy
##          0.835016             -0.102502
##      danceability             happiness
##          0.119070             0.017499
##      loudness_db             length_sec
##          1.147865             0.011877
##      age:energy             age:danceability
##          0.002489             -0.003376
##      age:loudness_db             raceWhite:danceability
##          -0.028004             -0.035957
##      raceWhite:length_sec             genderNot Male:genreOther
##          -0.014350             1.606214
##      genderNot Male:genrePop             genderNot Male:genreRock
##          0.660151             NA
##      genderNot Male:danceability             genderNot Male:loudness_db
##          -0.037786             -0.199855
##      genderNot Male:length_sec
##          -0.015603
```

```
model_2000s

##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##      danceability + happiness + bpm + length_sec + age:danceability +
##      race:genre + race:danceability + race:happiness + race:bpm +
##      race:length_sec + gender:danceability + gender:happiness +
##      gender:bpm, data = subpop)
##
## Coefficients:
##          (Intercept)                age
##          14.254799             -0.262392
##      genderNot Male             raceWhite
##          -0.551642             -13.479191
##      genreOther             genrePop
##          0.362481             0.218854
##      genreRock             energy
```



```
##           0.090807           0.018882
##           danceability       happiness
##           -0.083852           0.001354
##           bpm                length_sec
##           -0.014549          -0.009048
##           age:danceability    raceWhite:genreOther
##           0.003737           -1.186109
##           raceWhite:genrePop   raceWhite:genreRock
##           NA                  NA
##           raceWhite:danceability raceWhite:happiness
##           0.084078           -0.025028
##           raceWhite:bpm        raceWhite:length_sec
##           0.035295           0.019703
## genderNot Male:danceability    genderNot Male:happiness
##           -0.041970           0.021226
##           genderNot Male:bpm
##           0.019019
```

```
model_2010s
```

```
##
## Call:
## lm(formula = rating ~ age + gender + genre + energy + happiness +
##     bpm + length_sec + age:genre + age:bpm + age:length_sec +
##     gender:genre + gender:energy + gender:happiness, data = subpop)
##
## Coefficients:
##           (Intercept)                age          genderNot Male
##           -0.747913              0.207885          -1.434561
##           genrePop              genreRock              energy
##           4.006725             -1.384117          -0.034629
##           happiness                bpm          length_sec
##           0.017617             -0.050432           0.061106
##           age:genrePop          age:genreRock          age:bpm
##           -0.164148              0.066506           0.001726
##           age:length_sec  genderNot Male:genrePop  genderNot Male:genreRock
##           -0.001760              1.574766              NA
##           genderNot Male:energy  genderNot Male:happiness
##           0.040839             -0.027183
```

```
anova(model_results[["1960s"]] $\$$ first, model_60s)
```

```
## Analysis of Variance Table
```

```
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##     happiness + length_sec + race:genre + race:energy + race:danceability +
##     gender:danceability + gender:happiness
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     183 673.25
## 2     178 575.01  5     98.245 6.0826 3.166e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_60s, model_results[["1960s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##   happiness + length_sec + race:genre + race:energy + race:danceability +
##   gender:danceability + gender:happiness
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      178 575.01
## 2      155 535.98 23    39.027 0.4907 0.9759
```

```
anova(model_results[["1970s"]])$first, model_70s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##   age:length_sec + race:energy + gender:length_sec
##   Res.Df    RSS Df Sum of Sq    F  Pr(>F)
## 1       231 764.63
## 2       224 703.08  7    61.547 2.8012 0.008172 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_70s, model_results[["1970s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + happiness + bpm +
##   loudness_db + length_sec + age:genre + age:happiness + age:bpm +
##   age:length_sec + race:energy + gender:length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1       224 703.08
## 2       204 666.81 20    36.266 0.5547 0.9388
```

```
anova(model_results[["1980s"]])$first, model_80s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
```

```
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ race + genre + bpm + length_sec + race:genre + race:bpm +
##      race:length_sec
##   Res.Df    RSS Df Sum of Sq F Pr(>F)
## 1      216 494.88
## 2      218 485.57 -2      9.3015
```

```
anova(model_80s, model_results[["1980s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ race + genre + bpm + length_sec + race:genre + race:bpm +
##      race:length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      218 485.57
## 2      192 459.91 26      25.667 0.4121 0.9953
```

```
anova(model_results[["1990s"]])$first, model_90s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:energy + age:danceability +
##      age:loudness_db + race:danceability + race:length_sec + gender:genre +
##      gender:danceability + gender:loudness_db + gender:length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      122 269.95
## 2      113 230.77  9      39.186 2.132 0.03227 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_90s, model_results[["1990s"]])$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##      happiness + loudness_db + length_sec + age:energy + age:danceability +
##      age:loudness_db + race:danceability + race:length_sec + gender:genre +
##      gender:danceability + gender:loudness_db + gender:length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
```

```
## 1    113 230.77
## 2     98 217.50 15    13.262 0.3984 0.9766
anova(model_results[["2000s"]] $\$$ first, model_2000s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + length_sec + age:danceability + race:genre +
##   race:danceability + race:happiness + race:bpm + race:length_sec +
##   gender:danceability + gender:happiness + gender:bpm
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     116 270.33
## 2     108 208.61  8     61.719 3.994 0.0003515 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(model_2000s, model_results[["2000s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + length_sec + age:danceability + race:genre +
##   race:danceability + race:happiness + race:bpm + race:length_sec +
##   gender:danceability + gender:happiness + gender:bpm
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     108 208.61
## 2      93 194.20 15     14.415 0.4602 0.9544

anova(model_results[["2010s"]] $\$$ first, model_2010s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##   age:genre + age:bpm + age:length_sec + gender:genre + gender:energy +
##   gender:happiness
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     103 274.94
## 2      99 222.79  4     52.151 5.7935 0.0003127 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(model_2010s, model_results[["2010s"]] $\$$ second)

## Analysis of Variance Table
##
```

```

## Model 1: rating ~ age + gender + genre + energy + happiness + bpm + length_sec +
##   age:genre + age:bpm + age:length_sec + gender:genre + gender:energy +
##   gender:happiness
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      99 222.79
## 2      81 206.34 18    16.446 0.3587 0.9916

```